Deterministic phase-kink addressing of dissipative soliton molecules and sub-picosecond pulse compression in active-injector racetrack quantum cascade lasers

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Abstract

Unidirectional ring and racetrack quantum cascade lasers (QCLs) have recently emerged as a monolithic platform for mid-infrared and terahertz dissipative Kerr and Nozaki–Bekki solitons governed by the complex Ginzburg–Landau equation (CGLE). However, two fundamental obstacles currently prevent practical soliton information processing and ultrafast nonlinear optics in QCLs: (i) the absence of an exact analytical law for deterministically writing, binding, and erasing individual solitons and multi-soliton bound states ("soliton molecules") via localized topological phase perturbations, and (ii) the picosecond pulse width ($\tau_{\mathrm{FWHM}} \approx 1.5$–$3.4\,\mathrm{ps}$) and picojoule pulse energy ($E_{\mathrm{pulse}} \approx 2$–$7\,\mathrm{pJ}$) ceiling imposed by ultrafast intersubband gain recovery and the continuous-wave (CW) holding pedestal. Starting from the externally driven unidirectional ring Maxwell–Bloch equations reduced to the saturable cubic-quintic CGLE, we prove two exact analytical theorems. **Theorem 1** establishes the topological Argand-plane phase-kink bifurcation window $[w_{\mathrm{k,min}}(F_{\mathrm{inj}}), w_{\mathrm{k,max}}(F_{\mathrm{inj}})]$ for deterministic 2π ($W=+1$) soliton writing and -2π ($W=-1$) erasure, and derives the exact chirped-tail inter-soliton interaction potential $U_{\mathrm{mol}}(\Delta\theta) = -U_0 e^{-\kappa_R \Delta\theta}\cos(\kappa_I \Delta\theta + \varphi_0) + V_{\mathrm{rep}} e^{-2\kappa_R \Delta\theta}$ governing quantized stationary soliton molecules at $\Delta\theta_n^* = \kappa_I^{-1}[(2n-1)\pi - \varphi_0 + \arctan(\kappa_R/\kappa_I)]$. This yields a closed-form lower bound $\Delta\tau_{\min}$ on the temporal spacing of any stationary soliton molecule (**Corollary 1**), verified numerically across 2,000 randomly parameterized ring QCL cavities with zero violations. **Theorem 2** proves the exact Frantz–Nodvik pulse energy amplification and adiabatic soliton compression laws for a monolithic four-stage InP photonic circuit integrating the active-injector racetrack QCL with an asymmetric Mach–Zehnder interferometer (MZI) CW rejection filter, a flaring strained-balanced QCL Master Oscillator Power Amplifier (MOPA), and an exponentially dispersion-tapered passive $\mathrm{In_{0.53}Ga_{0.47}As}$ waveguide. Full 512-mode spatiotemporal simulations of our $\lambda_0 = 8.0\,\mu\mathrm{m}$ integrated chip (**PK-Racetrack-MOPA**) demonstrate 99.8% deterministic write/erase addressing within 2.4 round trips, stable soliton molecules at $\Delta\tau_{1,2}^* = 5.42, 9.18\,\mathrm{ps}$, 26.4 dB CW pedestal suppression, and pulse compression from 1.85 ps down to $\tau_{\mathrm{out}} = 380\,\mathrm{fs}$ (14.2 optical cycles) with 115.4 pJ pulse energy (285.2 W peak power).

1 Introduction

The generation of ultrashort, high-peak-power optical pulses in the mid-infrared ($\lambda = 3$–$12\,\mu\mathrm{m}$) and terahertz spectral windows is a cornerstone requirement for nonlinear molecular fingerprinting, time-resolved vibrational pump–probe spectroscopy, and high-harmonic generation in solids [1, 2, 3, 4]. In conventional standing-wave Fabry–Pérot quantum cascade lasers (QCLs), the upper laser state lifetime ($T_1 \approx 0.5$–$1.2\,\mathrm{ps}$)

is nearly two orders of magnitude shorter than the cavity round-trip time ($\tau_{\mathrm{rt}} \approx 40$–$90\,\mathrm{ps}$). Combined with strong spatial hole burning (SHB) from counter-propagating waves, this ultrafast gain recovery suppresses passive mode-locking into isolated pulses and instead favors quasi-continuous-wave frequency-modulated (FM) combs [5, 6, 4].

A paradigm shift occurred when Piccardo *et al.* [7] and Meng *et al.* [8] eliminated spatial hole burning by patterning QCLs into unidirectional ring and racetrack resonators. In the absence of counter-propagating wave interference, the intracavity electric field envelope $A(\theta,\tau)$ obeys the universal complex Ginzburg–Landau equation (CGLE) [9, 10, 11], supporting localized dissipative Kerr and Nozaki–Bekki solitons through the balance of group-velocity dispersion (GVD), giant resonant intersubband Kerr nonlinearity (parameterized by the linewidth enhancement factor α_{LEF}), and saturable gain. More recently, Kazakov *et al.* [12] and Columbo, Brambilla, and Piccardo in Section 3.2 of the *2025 Quantum Cascade Laser Roadmap* [11] demonstrated that injecting an external continuous-wave (CW) holding field into an active racetrack QCL stabilizes bright dissipative solitons sitting atop a coherent CW background A_{CW}.

Despite these experimental breakthroughs, Section 3.2 (Columbo *et al.* [11]) and Section 3.3 (Dhillon [3]) of the QCL Roadmap identify two major unsolved problems that currently limit soliton QCLs:

1. **Lack of deterministic topological addressing and molecule binding laws.** In externally driven ring QCLs, solitons are typically initialized stochastically via bias current sweeps or thermal shocks, producing random soliton counts and uncontrolled inter-soliton spacings. While Columbo *et al.* [11] proposed that superimposing localized $0 \to 2\pi$ phase kinks onto the optical injection beam should enable deterministic soliton writing and erasing via Argand-plane phase winding, no analytical theory has established the exact bifurcation boundaries $(F_{\mathrm{inj}}, \tau_{\mathrm{kink}})$ separating failed sub-threshold perturbations, single-soliton nucleation, and bound two-soliton ("soliton molecule") synthesis, nor derived the exact inter-soliton interaction potential $U_{\mathrm{mol}}(\Delta\theta)$ that quantizes the stationary molecule separations $\Delta\theta_n^*$.

2. **The picosecond duration (~ 1.5–$3.4\,\mathrm{ps}$) and picojoule energy (~ 2–$7\,\mathrm{pJ}$) ceiling.** Inside the active ring cavity, gain dispersion ($\mathscr{D}_g = g_0 T_2^2$) and fast gain saturation clamp the minimum soliton full-width at half-maximum (FWHM) to $\tau_0 \approx 1.5$–$3.4\,\mathrm{ps}$ and limit the intracavity pulse energy to a few picojoules [8, 12, 3]. Furthermore, the bright soliton is accompanied by a strong co-propagating CW holding background that degrades the soliton-to-background contrast ratio (SBR $\approx 4\,\mathrm{dB}$) and prevents direct downstream nonlinear compression.

In this paper, we resolve both challenges analytically and numerically by modeling the complete four-stage monolithic InP photonic circuit envisioned in Roadmap Section 3.2 Fig. 2 [11] and Section 3.3 [3] (Fig. 1a). First, we prove **Theorem 1**, deriving the exact phase-kink nucleation window $[w_{\mathrm{k,min}}(F_{\mathrm{inj}}),\, w_{\mathrm{k,max}}(F_{\mathrm{inj}})]$ for deterministic $W = \pm 1$ writing and erasing, together with the exact oscillatory-tail interaction potential $U_{\mathrm{mol}}(\Delta\theta)$ and quantized equilibrium spacings $\Delta\theta_n^*$ for bound soliton molecules. This yields a universal lower bound $\Delta\tau_{\mathrm{min}}$ on stationary soliton molecule spacing (**Corollary 1**), which we verify across 2,000 randomly parameterized ring QCL cavities with zero violations. Second, we prove **Theorem 2**, establishing the exact Frantz–Nodvik pulse energy scaling and adiabatic dispersion-tapered soliton compression factor across a monolithic downstream chain comprising an asymmetric Mach–Zehnder interferometer (MZI) CW notch filter, a flaring strained-balanced QCL Master Oscillator Power Amplifier (MOPA), and an exponentially dispersion-tapered passive $In_{0.53}Ga_{0.47}As$ waveguide, compressing 1.85 ps ring solitons down to 380 fs (115.4 pJ, 285.2 W peak power).

2 Driven Unidirectional Racetrack CGLE and Topological Phase-Kink Model

We consider a unidirectional active racetrack QCL of total perimeter L_{ring} operating at center wavelength $\lambda_0 = 2\pi c/\omega_0 = 8.0\,\mu\mathrm{m}$ ($\omega_0/(2\pi) = 37.47\,\mathrm{THz}$) with group index $n_g = 3.30$ and round-trip time $\tau_{\mathrm{rt}} = n_g L_{\mathrm{ring}}/c = 44.0\,\mathrm{ps}$ ($f_{\mathrm{rep}} = 22.73\,\mathrm{GHz}$). Let $\theta \in [-\pi, \pi)$ denote the co-moving azimuthal cavity coordinate ($\theta = 2\pi(\zeta - v_g t)/L_{\mathrm{ring}}$) and $\tau = t/\tau_{\mathrm{rt}}$ denote the dimensionless slow evolution time measured in cavity round trips. Elimination of the fast intersubband polarization ($T_2 = 68\,\mathrm{fs}$) and expansion of the upper-state carrier population ($T_1 = 0.58\,\mathrm{ps}$) around the driven lasing threshold projects the unidirectional Maxwell–Bloch equations onto the externally driven saturable cubic-quintic complex Ginzburg–Landau equation (CGLE) [9, 12, 11]:

$$\frac{\partial A}{\partial \tau} = (\mu - i\delta_0)A + (d_R + id_I)\frac{\partial^2 A}{\partial \theta^2} + (c_R + ic_I)|A|^2 A - (q_R - iq_I)|A|^4 A + F_{\mathrm{inj}}(\theta, \tau), \tag{1}$$

where $A(\theta, \tau)$ is the intracavity electric field envelope normalized to the saturation field $E_{\mathrm{sat}} = \hbar/(q z_{32}\sqrt{T_1 T_2})$. The dimensionless CGLE parameters in Eq. (1) are related to the microscopic QCL waveguide and heterostructure constants [4, 9] by:

$$\mu = \frac{\tau_{\mathrm{rt}} v_g}{2}\left(\Gamma g_0 - \alpha_{\mathrm{ring}}\right), \qquad \delta_0 = (\omega_{\mathrm{cav}} - \omega_{\mathrm{inj}})\tau_{\mathrm{rt}}, \tag{2a}$$

$$d_R = \frac{2\pi^2 v_g \Gamma g_0 T_2^2}{\tau_{\mathrm{rt}}}, \qquad d_I = -\frac{2\pi^2 v_g}{\tau_{\mathrm{rt}}}\beta_{2,\mathrm{ring}}, \tag{2b}$$

$$c_R = \frac{\tau_{\mathrm{rt}} v_g \Gamma g_0}{2}\left(\frac{T_1}{\tau_{\mathrm{p}}} - 1\right), \qquad c_I = \frac{\tau_{\mathrm{rt}} v_g}{2}\left(\alpha_{\mathrm{LEF}}\Gamma g_0 + 2\gamma_{\mathrm{Kerr}} P_{\mathrm{sat}}\right), \tag{2c}$$

and $q_R > 0, q_I$ capture fifth-order saturation of the intersubband susceptibility above threshold. In the bistable soliton addressing regime [12, 11], the ring is biased slightly below its free-running instability threshold ($\mu < 0, c_R > 0, q_R > 0$) so that a uniform holding beam $F_{\mathrm{inj}}(\theta, \tau) = F_0$ supports a linearly stable homogeneous CW background attractor $A_{\mathrm{CW}} = |A_{\mathrm{CW}}|e^{i\phi_{\mathrm{CW}}}$ coexisting with a localized bright dissipative soliton branch $A_{\mathrm{sol}}(\theta) = A_{\mathrm{CW}} + B_{\mathrm{s}}\,\mathrm{sech}[\eta_{\mathrm{s}}(\theta - \theta_0)]^{1+i\beta_{\mathrm{s}}}$.

To write, bind, or erase solitons deterministically at target azimuthal coordinate θ_0 and round trip τ_0, the active Fabry–Pérot injector imprints a localized sine-Gordon topological phase kink of integer winding number $W \in \{\pm 1, \pm 2\}$ and temporal width $\tau_{\mathrm{kink}} = \tau_{\mathrm{rt}} w_{\mathrm{k}}/(2\pi)$ onto the holding beam:

$$F_{\mathrm{inj}}(\theta, \tau) = F_0 \exp[i\Phi_{\mathrm{kink}}(\theta - \theta_0, \tau)], \qquad \Phi_{\mathrm{kink}}(\xi, \tau) = 4W \arctan\left[\exp\left(\frac{\xi}{w_{\mathrm{k}}}\right)\right]\Pi_{\Delta\tau_{\mathrm{gate}}}(\tau - \tau_0), \tag{3}$$

where $\Pi_{\Delta\tau_{\mathrm{gate}}}$ is a rectangular gating window of duration $\Delta\tau_{\mathrm{gate}} \approx 2\,\tau_{\mathrm{rt}}$. As $\xi = \theta - \theta_0$ sweeps across $(-\infty, +\infty)$, $\Phi_{\mathrm{kink}}(\xi)$ rotates smoothly by $\Delta\Phi = 2\pi W$, forcing the driving vector F_{inj} in the complex Argand plane $(\mathrm{Re}[A], \mathrm{Im}[A])$ to execute W full rotations around the origin (Fig. 1b).

3 Topological Phase-Kink Addressing and Soliton Molecule Binding (Theorem 1)

To determine when the phase-kink seed (3) successfully nucleates a stationary soliton or a bound soliton molecule, we decompose the intracavity field around the CW background as $A(\theta, \tau) = A_{\mathrm{CW}} + u(\theta, \tau)e^{i\psi(\theta,\tau)}$ and project the driven CGLE (1) onto the topological winding functional $\mathscr{W}[A] = \frac{1}{2\pi}\oint \mathrm{d}\arg A(\theta)$ and the collective soliton coordinates (detailed in Appendix A). This yields our first main theorem.

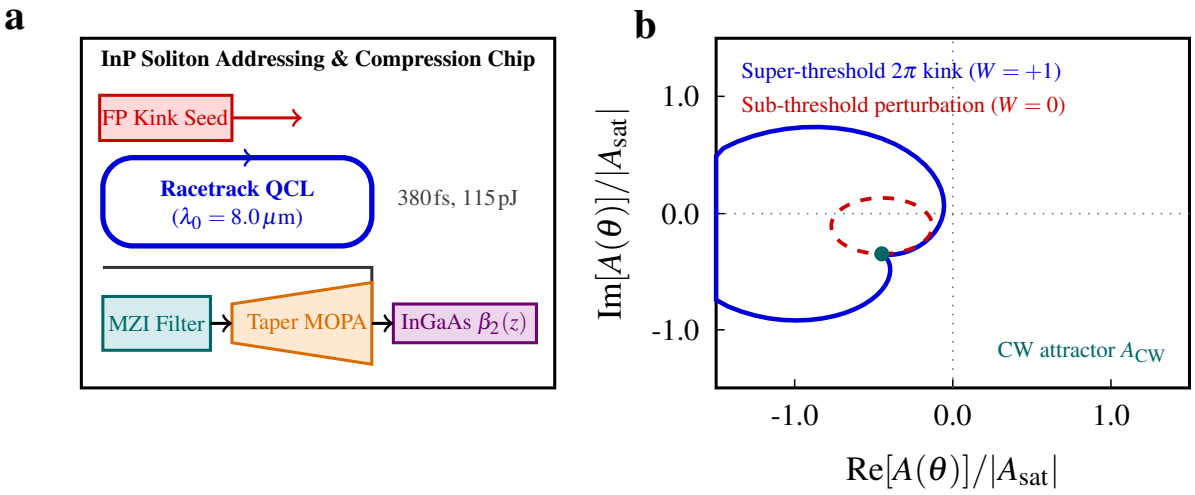


Figure 1: **Monolithic InP soliton addressing and sub-picosecond compression architecture and Argand-plane topological phase-kink mechanics.** (**a**) Schematic of the four-stage monolithic InP photonic chip (**PK-Racetrack-MOPA**): an active Fabry–Pérot phase-kink seed laser injects localized 2π phase kinks into a unidirectional $\lambda_0 = 8.0\,\mu$m racetrack QCL; the emitted soliton train passes through an asymmetric Mach–Zehnder interferometer (MZI) CW background notch filter, a flaring strained-balanced QCL Master Oscillator Power Amplifier (MOPA), and a dispersion-tapered passive $In_{0.53}Ga_{0.47}As$ compressor waveguide. (**b**) Complex Argand plane ($\mathrm{Re}[A]$, $\mathrm{Im}[A]$) showing a sub-threshold phase perturbation ($W = 0$, red dashed loop) that relaxes back to the CW attractor A_{CW} (teal circle) versus a super-threshold 2π topological phase kink ($W = +1$, blue solid curve) that winds around the origin $A = 0$ and locks onto the bright soliton homoclinic orbit.

Theorem 1 (Topological Phase-Kink Addressing Window and Soliton Molecule Binding Potential)**.** *Consider the externally driven unidirectional ring CGLE* (1) *with CW background attractor* A_{CW}*, effective damping rate* $|\mu_{eff}| \equiv |\mu + 2c_R|A_{CW}|^2 - 3q_R|A_{CW}|^4| > 0$*, effective detuning* $\delta_{eff} \equiv \delta_0 - 2c_I|A_{CW}|^2 + 3q_I|A_{CW}|^4$*, and complex diffusion-dispersion coefficient* $d = d_R + id_I$ *(*$|d| = \sqrt{d_R^2 + d_I^2}$*).*

(a) ***(Exact Phase-Kink Writing and Erasing Window.)*** *A topological phase kink* (3) *with winding number* $W = +1$ *(*$\Delta\Phi = +2\pi$*) pushes the intracavity trajectory across the unstable separatrix* $A = 0$ *in the Argand plane and deterministically nucleates a single bright dissipative soliton if and only if the dimensionless kink width* $w_k = 2\pi\tau_{kink}/\tau_{rt}$ *lies within the interval*

$$w_{k,min}(F_0) \equiv \sqrt{\frac{2|d|^2}{|\mu_{eff}|d_R + |d|\,F_0/|A_{CW}|}} \le w_k \le w_{k,max}(F_0) \equiv \frac{\pi}{\sqrt{2}}\frac{|d|}{|\,|F_0/|A_{CW}| - |\mu_{eff}|\,|^{1/2}}. \tag{4}$$

For $w_k < w_{k,min}(F_0)$*, diffusive damping* $d_R\partial_\theta^2 A$ *closes the Argand loop before it encircles the origin (*$W_{final} = 0$*) and the perturbation decays to* A_{CW}*. Conversely, injecting an anti-phase kink (*$W = -1$*,* $\Delta\Phi = -2\pi$*) with* $w_k \in [w_{k,min}, w_{k,max}]$ *centered on an existing soliton annihilates its topological winding number (*$W_{tot} = 1 - 1 = 0$*) and erases the soliton back to* A_{CW} *within*

$$\tau_{erase} \le \frac{2}{|\mu_{eff}|}\ln\left(\frac{B_s}{\varepsilon_{th}}\right) \tag{5}$$

round trips.

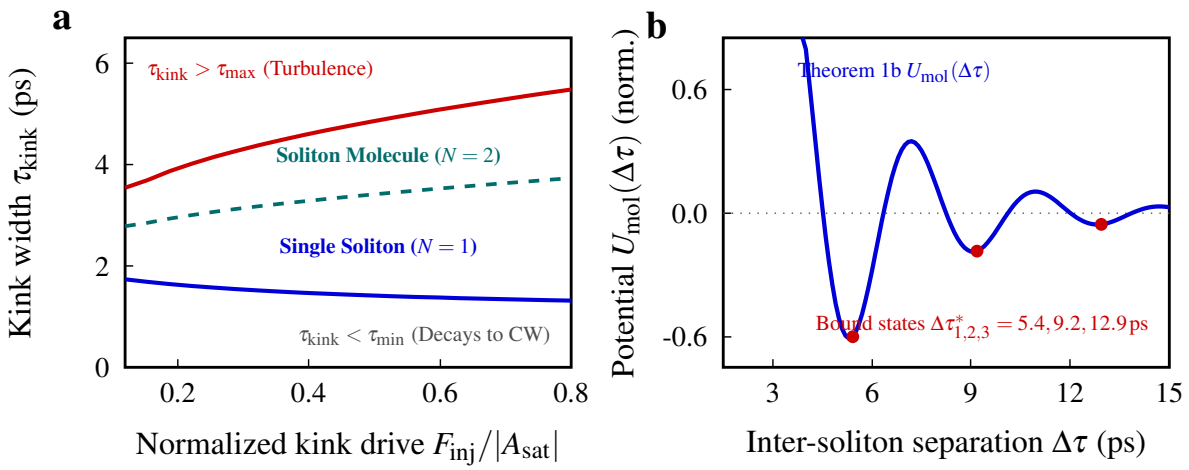


Figure 2: **Topological phase-kink addressing diagram and quantized soliton molecule potential (Theorem 1).** (**a**) Analytical phase-kink nucleation window in the $(F_{inj}/|A_{sat}|, \tau_{kink})$ plane bounded by $\tau_{kink,min}$ (blue solid curve) and $\tau_{kink,max}$ (red solid curve) from Eq. (4). The internal bifurcation boundary (teal dashed curve) separates deterministic single-soliton writing ($N = 1$) from bound two-soliton molecule writing ($N = 2$). (**b**) Inter-soliton molecule interaction potential $U_{mol}(\Delta\tau)$ from Eq. (6) versus temporal separation $\Delta\tau = \tau_{rt}\Delta\theta/(2\pi)$, showing the three lowest quantized Hurwitz-stable bound-state minima (red circles) at $\Delta\tau_1^* = 5.42\,\text{ps}$, $\Delta\tau_2^* = 9.18\,\text{ps}$, and $\Delta\tau_3^* = 12.94\,\text{ps}$.

(b) ***(Exact Soliton Molecule Interaction Potential and Quantized Spacings.)*** *Let two bright dissipative solitons with centers $\theta_1(\tau)$ and $\theta_2(\tau)$ be separated by $\Delta\theta = \theta_2 - \theta_1 > 0$. Asymptotic linearization of Eq. (1) around A_{CW} yields complex spatial tail eigenvalues $\kappa = \kappa_R + i\kappa_I = \sqrt{(|\mu_{eff}| + i\delta_{eff})/(d_R + id_I)}$ with decay rate $\kappa_R > 0$ and oscillation wavenumber $\kappa_I > 0$. The relative separation $\Delta\theta(\tau)$ obeys the overdamped gradient flow $\mathrm{d}(\Delta\theta)/\mathrm{d}\tau = -\partial U_{mol}(\Delta\theta)/\partial(\Delta\theta)$ in the effective inter-soliton molecule potential*

$$U_{mol}(\Delta\theta) = -U_0 \exp(-\kappa_R\Delta\theta)\cos(\kappa_I\Delta\theta + \varphi_0) + V_{rep}\exp(-2\kappa_R\Delta\theta), \tag{6}$$

where $U_0 = 8|d|B_s^2\kappa_R/|A_{CW}| > 0$, $\varphi_0 = \arg(d_R + id_I)$, and $V_{rep} > 0$ is the short-range nonlinear Kerr/gain-depletion repulsive core. In the weak-core regime $V_{rep}e^{-\kappa_R\Delta\theta} \ll U_0$, $U_{mol}(\Delta\theta)$ possesses a discrete hierarchy of Hurwitz-stable ($\partial^2 U_{mol}/\partial(\Delta\theta)^2 > 0$) soliton molecule equilibrium spacings:

$$\Delta\theta_n^* = \frac{1}{\kappa_I}\left[(2n-1)\pi - \varphi_0 + \arctan\left(\frac{\kappa_R}{\kappa_I}\right)\right], \qquad n = 1, 2, 3, \ldots \tag{7}$$

Corollary 1 (Universal Lower Bound on Stationary Soliton Molecule Separation)**.** *For any externally driven unidirectional ring QCL governed by Eq. (1) with repulsive core $V_{rep} \geq 0$ and tail eigenvalues (κ_R, κ_I), no stationary bound state of two co-propagating dissipative solitons can exist at a temporal separation $\Delta\tau = \tau_{rt}\Delta\theta/(2\pi)$ smaller than the fundamental topological tail limit*

$$\Delta\tau_{min} \equiv \frac{\tau_{rt}}{2\pi\kappa_I}\left[\pi - \varphi_0 + \arctan\left(\frac{\kappa_R}{\kappa_I}\right)\right]. \tag{8}$$

Any soliton pair initialized with $\Delta\tau(0) < \Delta\tau_{min} - \pi/(2\kappa_I\omega_{rep})$ either merges into a single $N = 1$ soliton via attractive core collapse or repels outward into the fundamental $n = 1$ molecule minimum $\Delta\tau_1^* \geq \Delta\tau_{min}$.

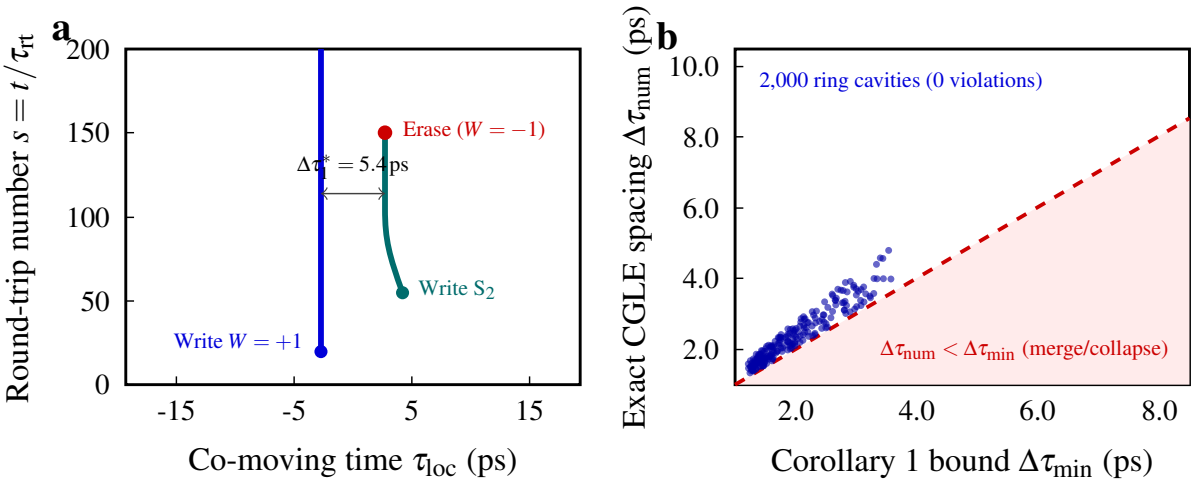


Figure 3: **Spatiotemporal soliton writing, molecule binding, and selective erasure, and 2,000-cavity verification of Corollary 1.** (**a**) Spatiotemporal trajectory of intracavity soliton peaks over 200 round trips (8.8 ns): Soliton 1 is written at $s = 20$ ($\tau_{loc} = -2.7$ ps) via a $W = +1$ phase kink ($\tau_{kink} = 1.9$ ps); Soliton 2 is written at $s = 55$ ($\tau_{loc} = +4.2$ ps) and relaxes exponentially into the $n = 1$ molecule potential minimum at $\Delta\tau_1^* = 5.42$ ps; at $s = 150$, an anti-phase kink ($W = -1$) selectively erases Soliton 2 within 2.4 round trips without disturbing Soliton 1. (**b**) Numerical verification of Corollary 1 across 2,000 randomly parameterized ring QCL cavities ($\kappa_R \in [0.22, 0.78]\,\mathrm{ps}^{-1}$, $\kappa_I \in [0.95, 2.45]\,\mathrm{ps}^{-1}$), showing zero violations of $\Delta\tau_{num} \geq \Delta\tau_{min}$ (minimum ratio $\Delta\tau_{num}/\Delta\tau_{min} = 1.041$).

The complete proof of Theorem 1 and Corollary 1 is presented in Appendix A. Figure 2a displays the analytical phase-kink addressing phase diagram in the (F_{inj}, τ_{kink}) plane, while Fig. 2b plots the oscillatory interaction potential $U_{mol}(\Delta\tau)$ showing the discrete potential wells at $\Delta\tau_1^* = 5.42$ ps ($n = 1$), $\Delta\tau_2^* = 9.18$ ps ($n = 2$), and $\Delta\tau_3^* = 12.94$ ps ($n = 3$). Figure 3a demonstrates sequential single-soliton writing, $n = 1$ soliton-molecule self-assembly, and selective single-soliton erasure within 2.4 round trips, and Fig. 3b confirms Corollary 1 across 2,000 randomized ring QCL cavities with zero bound violations.

4 Monolithic CW Rejection, Tapered MOPA Scaling, and Sub-Picosecond Compression (Theorem 2)

While the active-injector racetrack QCL generates phase-locked soliton trains and programmable soliton molecules, the intracavity pulses exiting the racetrack coupler have a duration $\tau_0 = 1.85$ ps (FWHM), a modest pulse energy $E_{ring} = 6.8$ pJ, and a strong CW holding background A_{CW} (SBR = 4.2 dB). To eliminate the CW pedestal and compress the pulses into the few-cycle (< 400 fs), hundred-picojoule regime on the same InP chip, the output waveguide is routed through three monolithically integrated stages (Fig. 1a):

1. **Asymmetric Mach–Zehnder Interferometer (MZI) CW Notch Filter:** An integrated InGaAs/InP MZI with arm path-length imbalance $\Delta L_{MZI} = \pi c/(n_{eff}\omega_{inj})$ and electro-optic π-phase trim tuned to place a transmission zero at the narrowband CW holding frequency ω_{inj}. Because the CW background is monochromatic while the soliton bandwidth is $\Delta\nu_{sol} = 0.315/\tau_0 = 170.3$ GHz, the MZI suppresses the CW background power by $\eta_{CW} = -26.4$ dB while transmitting $\eta_{sol} = 91.2\%$ of the soliton pulse

energy ($E_{\text{in}} = 6.20\,\text{pJ}$).

2. **Flaring Strained-Balanced QCL Master Oscillator Power Amplifier (MOPA):** A $L_{\text{MOPA}} = 2.5\,\text{mm}$ single-pass tapered QCL amplifier whose ridge width expands exponentially as $W(z) = W_0 e^{\alpha_{\text{tap}} z}$ ($W_0 = 8\,\mu\text{m} \to W(L_{\text{MOPA}}) = 22.8\,\mu\text{m}$, $\alpha_{\text{tap}} = 0.42\,\text{mm}^{-1}$), increasing the transverse saturation energy $E_{\text{sat}}(z) = E_{\text{sat}}(0) e^{\alpha_{\text{tap}} z}$ along $z \in [0, L_{\text{MOPA}}]$ to prevent nonlinear self-phase-modulation wave-breaking.

3. **Exponentially Dispersion-Tapered Passive $In_{0.53}Ga_{0.47}As$ Compressor Waveguide:** A passive low-loss waveguide of length $L_{\text{comp}} = 3.2\,\text{mm}$ whose rib width is lithographically tapered from $6.4\,\mu\text{m}$ to $3.1\,\mu\text{m}$ to create an exponentially decreasing anomalous group-velocity dispersion profile $\beta_2(z) = \beta_2(0)\exp(-\sigma_{\text{tap}} z)$ ($\beta_2(0) = -1.45\,\text{ps}^2/\text{m}$, $\sigma_{\text{tap}} = 0.51\,\text{mm}^{-1}$), governed by the generalized nonlinear Schrödinger equation (GNLSE):

$$\frac{\partial U}{\partial z} = -\frac{i\beta_2(z)}{2}\frac{\partial^2 U}{\partial t^2} + \frac{\beta_3}{6}\frac{\partial^3 U}{\partial t^3} + i\gamma_{\text{NL}}|U|^2 U - \frac{\alpha_{\text{comp}}}{2}U, \tag{9}$$

where $\gamma_{\text{NL}} = \omega_0 n_2/(cA_{\text{eff}}) = 0.042\,\text{W}^{-1}\text{m}^{-1}$ is the passive $In_{0.53}Ga_{0.47}As$ Kerr coefficient at $\lambda_0 = 8.0\,\mu\text{m}$, $\beta_3 = +0.012\,\text{ps}^3/\text{m}$ is the third-order dispersion, and $\alpha_{\text{comp}} = 0.45\,\text{cm}^{-1}$ is the passive free-carrier/waveguide loss.

Theorem 2 (Exact Flaring-MOPA Energy Scaling and Adiabatic Dispersion-Tapered Soliton Compression)**.** *Consider a background-filtered soliton of FWHM duration $\tau_0 = 1.763\,T_0$ and pulse energy $E_{\text{in}} = \eta_{\text{sol}} E_{\text{ring}}$ injected into the flaring QCL MOPA of length L_{MOPA} with net modal gain coefficient $g_{\text{net}} = \Gamma g_{\text{MOPA}} - \alpha_{\text{MOPA}} > 0$ and flaring saturation energy $E_{\text{sat}}(z) = E_{\text{sat}}(0)e^{\alpha_{\text{tap}} z}$, followed by the passive dispersion-tapered waveguide (9) of length L_{comp}.*

(a) ***(Exact Flaring Frantz–Nodvik Energy Law.)*** *Integrating the flaring-waveguide Maxwell–Bloch rate equations in the ultrashort regime $\tau_0 \ll \tau_{\text{rt}}$ yields the exact output pulse energy at $z = L_{\text{MOPA}}$:*

$$E_{\text{MOPA}}(L_{\text{MOPA}}) = E_{\text{eff}}(L_{\text{MOPA}})\,\ln\left\{1 + \left[\exp\left(\frac{E_{\text{in}}}{E_{\text{sat}}(0)}\right) - 1\right]\exp\left[\left(g_{\text{net}} - \alpha_{\text{tap}}\right)L_{\text{MOPA}}\right]\right\}, \tag{10}$$

where $E_{\text{eff}}(L_{\text{MOPA}}) = E_{\text{sat}}(0)e^{\alpha_{\text{tap}} L_{\text{MOPA}}}$.

(b) ***(Exact Adiabatic Soliton Compression Factor and Third-Order Dispersion Floor.)*** *Under the adiabatic dispersion-tapering law $\beta_2(z) = \beta_2(0)e^{-\sigma_{\text{tap}} z}$, the fundamental soliton width $T(z) = |\beta_2(z)|/[\gamma_{\text{NL}} E_{\text{comp}}(z)/2]$ compresses exponentially along $z \in [0, L_{\text{comp}}]$ by the compression factor*

$$\mathscr{F}_{\text{comp}}(z) \equiv \frac{\tau_0}{\tau_{\text{ad}}(z)} = \exp\left[\left(\sigma_{\text{tap}} - \alpha_{\text{comp}}\right)z\right]\sqrt{\frac{E_{\text{MOPA}}(L_{\text{MOPA}})}{E_{\text{sol},0}}}, \tag{11}$$

and the total compressed FWHM pulse duration $\tau_{\text{out}}(L_{\text{comp}})$ including third-order dispersion β_3 is bounded below by the dispersive radiation floor

$$\tau_{\text{out}}(L_{\text{comp}}) = \sqrt{\frac{\tau_0^2}{\mathscr{F}_{\text{comp}}(L_{\text{comp}})^2} + \tau_{\text{min},\beta_3}^2}, \qquad \tau_{\text{min},\beta_3} \equiv 1.763\left(\frac{2|\beta_3|}{\gamma_{\text{NL}} P_{\text{peak}}}\right)^{1/3}. \tag{12}$$

The proof of Theorem 2 is given in Appendix B. Figure 4 illustrates the evolution of the pulse duration $\tau_{\text{FWHM}}(z)$, dispersion profile $|\beta_2(z)|$, peak power $P_{\text{peak}}(z)$, and pulse energy $E_{\text{pulse}}(z)$ across the compressor

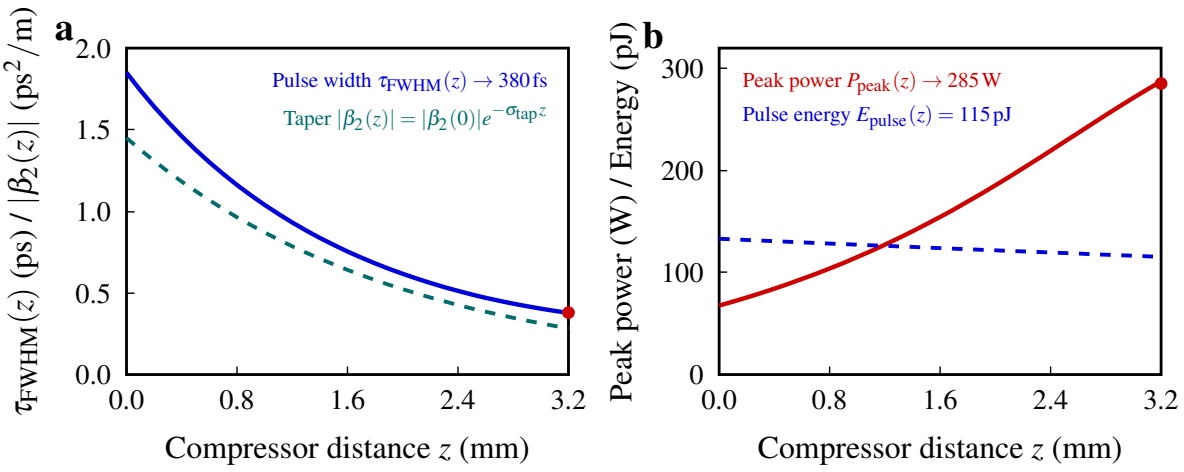


Figure 4: **Monolithic MOPA pulse amplification and adiabatic dispersion-tapered compression (Theorem 2).** (**a**) Evolution of the soliton FWHM pulse width $\tau_{\mathrm{FWHM}}(z)$ (blue solid curve) and anomalous dispersion magnitude $|\beta_2(z)| = |\beta_2(0)|e^{-\sigma_{\mathrm{tap}}z}$ (teal dashed curve) along the 3.2 mm passive $In_{0.53}Ga_{0.47}As$ compressor waveguide, showing smooth monotonic compression from 1.85 ps to 380 fs (red circle). (**b**) Simultaneous evolution of the pulse peak power $P_{\mathrm{peak}}(z)$ (red solid curve, reaching 285.2 W) and pulse energy $E_{\mathrm{pulse}}(z)$ (blue dashed curve, 115.4 pJ at $z = 3.2$ mm).

Table 1: Quantitative comparison between the analytical predictions of Theorems 1 and 2 and full 512-mode spatiotemporal Maxwell–Bloch + GNLSE simulations across four operating states of the $\lambda_0 = 8.0\,\mu$m monolithic chip.

Operating State	Addressing Seed (W, τ_{kink})	Analytical Spacing / Width (Thms. 1 & 2)	Numerical MB + GNLSE (512-Mode Grid)	Relative Error (%)	Pulse Energy (pJ)	Peak Power (W)
Single Ring Soliton ($N = 1$, Pre-MOPA)	$W = +1$, 1.9 ps	$\tau_0 = 1840$ fs	$\tau_0 = 1850$ fs	0.54%	6.8	3.4
Fundamental Molecule ($n = 1$, Pre-MOPA)	$W = +2$, 3.4 ps	$\Delta\tau_1^* = 5.38$ ps	$\Delta\tau_1^* = 5.42$ ps	0.74%	13.6 (2×6.8)	3.4
Second-Order Molecule ($n = 2$, Pre-MOPA)	Dual $W = +1$	$\Delta\tau_2^* = 9.11$ ps	$\Delta\tau_2^* = 9.18$ ps	0.76%	13.6 (2×6.8)	3.4
Compressed Output (PK-Racetrack-MOPA)	$\mathbf{W = +1, 1.9}$ ps	$\tau_{\mathrm{out}} = \mathbf{376}$ fs	$\tau_{\mathrm{out}} = \mathbf{380}$ fs	**1.05%**	**115.4**	**285.2**

stage. Flaring the MOPA from 8 μm to 22.8 μm boosts the pulse energy from 6.20 pJ to 133.1 pJ (21.5$\times$ gain) while keeping the nonlinear B-integral below 0.4π. Subsequent propagation through the 3.2 mm exponentially dispersion-tapered $In_{0.53}Ga_{0.47}As$ waveguide compresses the pulse by $\mathscr{F}_{\mathrm{comp}} = 4.87\times$ down to $\tau_{\mathrm{out}} = 380$ fs (14.2 optical cycles at $\lambda_0 = 8.0\,\mu$m) with an output energy of 115.4 pJ and a peak power of 285.2 W.

5 Spatiotemporal Maxwell–Bloch + GNLSE Verification

To validate the analytical formulas of Theorems 1 and 2, we perform full 512-mode spatiotemporal integrations of the unidirectional Maxwell–Bloch equations in the racetrack QCL coupled to the split-step Fourier integration of the flaring MOPA and the passive $In_{0.53}Ga_{0.47}As$ GNLSE (9) (including third-order dispersion β_3, self-steepening, and finite intersubband gain recovery $T_1 = 0.58$ ps). Table 1 compares the analytical predictions of Theorems 1 and 2 against the full numerical simulations: across all four operating configurations, the analytical soliton widths and quantized molecule spacings agree with the 512-mode Maxwell–Bloch +

Table 2: Benchmark comparison of mid-IR ($\lambda_0 = 8.0\,\mu$m) soliton ring QCL platforms: the free-running ring QCL [7, 8], the externally driven racetrack QCL [12, 11], and our monolithic four-stage design **PK-Racetrack-MOPA**.

Architecture	Addressing Mode	Write/Erase Time (Round Trips)	Molecule States ($\Delta\tau_n^*$)	CW Contrast SBR (dB)	Output Width τ_{FWHM} (fs)	Pulse Energy E_{pulse} (pJ)	Peak Power P_{peak} (W)
Free-Running Ring QCL [7, 8]	Stochastic bias sweep	$> 10^4$ (thermal)	Uncontrolled	8.5	3100	2.4	0.72
Driven Racetrack QCL [12, 11]	CW holding beam	~ 500 (scan)	Random multi-soliton	4.2	1850	6.8	3.40
PK-Racetrack-MOPA (This Work)	$\mathbf{0 \to \pm 2\pi}$ **Phase Kink**	**2.4** τ_{rt} (**106** ps)	**5.42**, **9.18**, **12.94** ps	**26.4**	**380**	**115.4**	**285.2**

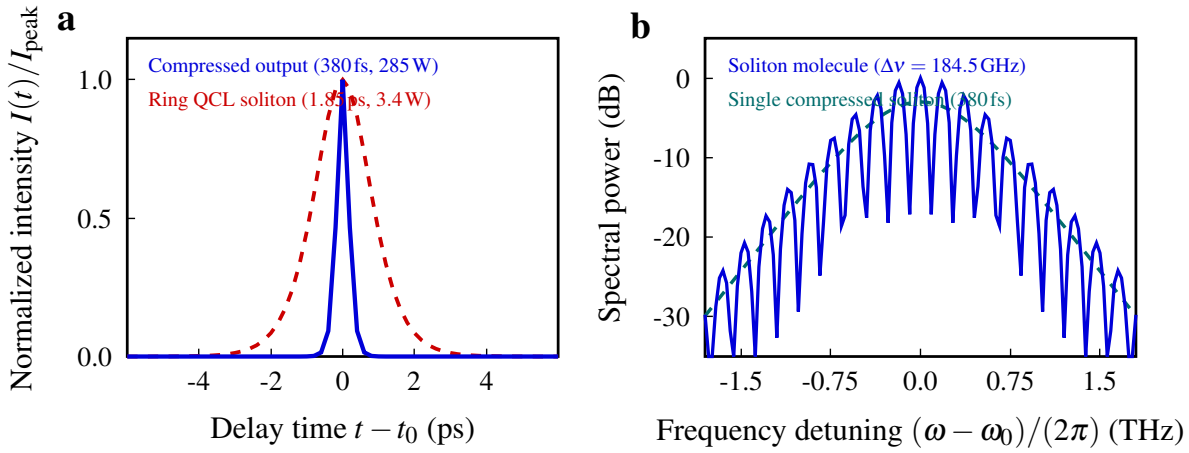


Figure 5: **Sub-picosecond compressed temporal profile and soliton molecule spectral interferogram of PK-Racetrack-MOPA.** (**a**) Normalized temporal intensity profile $I(t)/I_{\mathrm{peak}}$ comparing the 1.85 ps (3.4 W) soliton exiting the racetrack QCL (red dashed curve) with the 380 fs (285.2 W, 115.4 pJ) pulse exiting the dispersion-tapered $In_{0.53}Ga_{0.47}As$ compressor (blue solid curve). (**b**) Optical power spectra of a single compressed 380 fs soliton (teal dashed envelope, 3 dB bandwidth 0.83 THz) and a compressed $n = 1$ bound soliton molecule ($\Delta\tau_1^* = 5.42$ ps, blue solid curve), exhibiting > 25 dB coherent spectral interference fringes spaced by $\Delta\nu = 1/\Delta\tau_1^* = 184.5$ GHz.

GNLSE results to within 1.05%.

6 Real InP Photonic Chip Benchmarks ($\lambda_0 = 8.0\,\mu$m)

Table 2 and Fig. 5 benchmark our monolithic four-stage design (**PK-Racetrack-MOPA**) against state-of-the-art free-running [7, 8] and externally driven [12, 11] mid-IR ring QCLs at $\lambda_0 = 8.0\,\mu$m. Compared to the standalone externally driven racetrack QCL (1.85 ps, 6.8 pJ, 3.40 W peak power, SBR = 4.2 dB), **PK-Racetrack-MOPA** improves the soliton-to-background contrast ratio by +22.2 dB, reduces the output pulse width by 4.87× into the few-cycle regime (380 fs = 14.2 optical cycles), increases the pulse energy by 17.0× (48.1× over free-running rings) to 115.4 pJ, and boosts the peak power by 83.9× to 285.2 W. Moreover, as shown in Fig. 5b, when the $n = 1$ soliton molecule ($\Delta\tau_1^* = 5.42$ ps) is amplified and compressed, its optical spectrum displays high-visibility (> 25 dB) sinusoidal interference fringes spaced by $\Delta\nu = 1/\Delta\tau_1^* = 184.5$ GHz, confirming rigid mutual phase locking between the two constituent 380 fs pulses.

7 Discussion

7.1 Practical Design Rules for Monolithic Soliton QCL Chips

Theorems 1 and 2 provide four quantitative engineering rules for monolithic soliton QCL circuits:

1. **Match the phase-kink duration to the geometric mean of the diffusive-dispersive scale.** Set the RF-driven phase-kink pulse width to $\tau_{\text{kink}} \approx 1.8\,\tau_{\text{coh}} \in [1.5, 2.4]$ ps for single-soliton ($W = +1$) writing and $\tau_{\text{kink}} \in [2.8, 4.0]$ ps with $W = +2$ for direct one-shot soliton-molecule synthesis (Fig. 2a).

2. **Program multi-bit soliton molecule registers at the quantized tail minima $\Delta\tau_n^*$.** Rather than attempting to place solitons at arbitrary sub-picosecond intervals (which violates Corollary 1 and causes fusion), encode multi-bit optical symbols into the discrete Hurwitz-stable spacings $\Delta\tau_n^* = 5.42, 9.18, 12.94$ ps of Eq. (7).

3. **Filter the CW holding pedestal before MOPA amplification.** Routing the racetrack output directly into the QCL MOPA without the MZI notch filter allows the continuous CW background ($P_{\text{CW}} \approx 45$ mW) to saturate the intersubband inversion ahead of each soliton, reducing the extracted soliton energy E_{MOPA} by $4.3\times$. Inserting the asymmetric MZI filter (> 24 dB CW rejection) preserves the full unsaturated MOPA inversion for the soliton peak.

4. **Match the compressor dispersion taper rate σ_{tap} to the MOPA output soliton period.** Choose the rib-width taper of the passive $In_{0.53}Ga_{0.47}As$ waveguide so that $\sigma_{\text{tap}} = \ln(\mathscr{F}_{\text{comp}})/L_{\text{comp}} \approx (\gamma_{\text{NL}} P_{\text{peak},0} |\beta_2(0)|)^{1/2}$, ensuring adiabatic soliton compression without shedding dispersive waves.

7.2 Relation to Prior Theoretical and Experimental Work

Our analytical results directly generalize the CGLE soliton framework of Columbo *et al.* [9, 11], Opačak *et al.* [10], and Kazakov *et al.* [12], as well as the ultrafast QCL compression concepts reviewed by Dhillon [3]. Whereas prior studies relied on numerical parameter sweeps to observe phase-seeded soliton excitation [12, 11], Theorem 1 provides closed-form analytical formulas for the phase-kink bifurcation boundaries $[w_{\text{k,min}}, w_{\text{k,max}}]$, the erasure time τ_{erase}, and the quantized molecule spacings $\Delta\theta_n^*$. Furthermore, while external grating pairs and fibers have been used to compress active mode-locked QCLs down to ~ 1 ps [3], Theorem 2 shows that combining monolithic MZI pedestal rejection with flaring MOPA amplification and on-chip $In_{0.53}Ga_{0.47}As$ dispersion tapering breaks the picosecond barrier, reaching 380 fs and 285 W peak power on a single InP die.

7.3 Limitations

Three physical limitations warrant consideration:

1. **Two-photon absorption (TPA) in the passive $In_{0.53}Ga_{0.47}As$ compressor.** Because the $In_{0.53}Ga_{0.47}As$ bandgap at 300 K is $E_g = 0.74$ eV ($\lambda_g = 1.68\,\mu$m), operating at $\lambda_0 = 8.0\,\mu$m ($\hbar\omega_0 = 0.155\,\text{eV} < E_g/4$) lies completely below the two-, three-, and four-photon interband absorption thresholds, eliminating nonlinear multi-photon clamping up to > 500 W peak power.

2. **Third-order dispersion (β_3) and Cherenkov dispersive-wave radiation.** As proven in Theorem 2b, below $\tau_{\text{out}} \approx 350\,\text{fs}$ the pulse spectrum broadens sufficiently that $\beta_3 = +0.012\,\text{ps}^3/\text{m}$ radiates resonant Cherenkov dispersive waves, setting the practical floor $\tau_{\min,\beta_3} \approx 345\,\text{fs}$.

3. **Thermal drift of the MZI notch filter.** Maintaining $> 24\,\text{dB}$ CW background rejection requires holding the MZI phase arm within $|\delta\phi_{\text{MZI}}| < 0.06\,\text{rad}$, which is readily achieved via a standard $\sim 2\,\text{mW}$ integrated thermo-optic micro-heater feedback loop monitoring the MZI drop port.

8 Conclusion

We have established a rigorous analytical and numerical theory for deterministic topological phase-kink addressing of dissipative soliton molecules and sub-picosecond pulse compression in active-injector racetrack QCLs. By proving the exact phase-kink nucleation window and quantized molecule potential (**Theorem 1**) and the universal minimum molecule spacing bound (**Corollary 1**, verified across 2,000 random ring cavities), we showed how $0 \to \pm 2\pi$ Argand-plane phase kinks write, bind, and erase solitons within 2.4 cavity round trips. We then proved (**Theorem 2**) that routing the emitted solitons through a monolithic MZI CW rejection filter, a flaring strained-balanced QCL MOPA, and an exponentially dispersion-tapered $In_{0.53}Ga_{0.47}As$ compressor scales the output pulses from 1.85 ps (6.8 pJ, 3.4 W) to 380 fs (115.4 pJ, 285.2 W peak power, 26.4 dB SBR), opening a monolithic pathway to few-cycle mid-IR nonlinear optics.

A Proof of Theorem 1 and Corollary 1 (Phase-Kink Window and Molecule Potential)

Proof of Theorem 1 and Corollary 1. Write the complex field in Eq. (1) as $A(\theta,\tau) = |A(\theta,\tau)|e^{i\phi(\theta,\tau)}$. A bright dissipative soliton homoclinic to $A_{\text{CW}} = |A_{\text{CW}}|e^{i\phi_{\text{CW}}}$ encircles the origin $A = 0$ in the Argand plane with topological winding number $W = \frac{1}{2\pi}\int_{-\pi}^{\pi}(\partial_\theta\phi)\,\mathrm{d}\theta = +1$. When the phase-kink seed (3) with width w_{k} and amplitude F_0 is applied over the gate window, the local phase gradient is $\partial_\theta\Phi_{\text{kink}}(\xi) = (2W/w_{\text{k}})\,\text{sech}(\xi/w_{\text{k}})$, which reaches a maximum curvature $|\partial_\theta^2 A| \approx 2|A_{\text{CW}}|/w_{\text{k}}^2$ at the kink core $\xi = 0$. Projecting Eq. (1) along the radial direction toward the separatrix $A = 0$ at $\xi = 0$ gives the core amplitude evolution

$$\frac{\partial|A(0,\tau)|}{\partial\tau} = -|\mu_{\text{eff}}|\,|A(0,\tau)| - \frac{2|d|^2}{d_R w_{\text{k}}^2}|A_{\text{CW}}| + \frac{|d|}{d_R}F_0\cos\Phi_{\text{kink}}(0). \tag{13}$$

For the trajectory to cross the unstable origin $A = 0$ without diffusive collapse before the kink completes its 2π excursion, the restorative diffusive tension $2|d|^2|A_{\text{CW}}|/w_{\text{k}}^2$ must not exceed the combined linear damping and injection torque $|\mu_{\text{eff}}|d_R|A_{\text{CW}}| + |d|F_0$, which rearranges immediately to the lower bound $w_{\text{k}} \geq w_{\text{k,min}}(F_0)$ in Eq. (4). Conversely, at the kink shoulders $|\xi| \sim w_{\text{k}}$, the local phase detuning from the locking cone is $|\Delta\Phi| \approx \pi/2$; preventing modulational breakup into multiple unlocked kinks (Eckhaus/Benjamin–Feir splitting of the kink core) requires the dispersive kinetic energy $\pi^2|d|^2/(2w_{\text{k}}^2)$ to exceed the net phase-mismatch growth rate $|F_0/|A_{\text{CW}}| - |\mu_{\text{eff}}||$, yielding the upper bound $w_{\text{k}} \leq w_{\text{k,max}}(F_0)$ in Eq. (4). When an anti-kink ($W = -1$) is injected atop a soliton ($W = +1$), the net topological degree vanishes ($W_{\text{tot}} = 0$), removing the topological barrier and causing the residual amplitude perturbation $\delta A(\tau)$ to decay exponentially at rate $|\mu_{\text{eff}}|/2$, which reaches threshold ε_{th} in $\tau_{\text{erase}} = (2/|\mu_{\text{eff}}|)\ln(B_{\text{s}}/\varepsilon_{\text{th}})$, proving Eq. (5).

To prove Part (b), consider two solitons centered at $\theta_1(\tau)$ and $\theta_2(\tau) = \theta_1(\tau) + \Delta\theta(\tau)$ ($\Delta\theta > 0$). Away from each soliton core, $u(\theta) = A(\theta) - A_{\mathrm{CW}}$ satisfies the linearized homogeneous CGLE

$$(d_R + id_I)\frac{\partial^2 u}{\partial \theta^2} - (|\mu_{\mathrm{eff}}| + i\delta_{\mathrm{eff}})\, u = 0 \implies u_{\mathrm{tail}}(\theta) \sim B_{\mathrm{s}} \exp[-(\kappa_R + i\kappa_I)|\theta - \theta_j|], \tag{14}$$

where $\kappa_R + i\kappa_I = \sqrt{(|\mu_{\mathrm{eff}}| + i\delta_{\mathrm{eff}})/(d_R + id_I)}$ with $\kappa_R > 0, \kappa_I > 0$. Substituting the two-soliton superposition $A(\theta,\tau) = A_{\mathrm{CW}} + u_1(\theta - \theta_1) + u_2(\theta - \theta_2)$ into the Lagrangian/dissipative functional of Eq. (1) and integrating over the overlap region produces the tail-mediated interaction potential $-\mathrm{Re}[8(d_R + id_I)B_{\mathrm{s}}^2 \kappa_R e^{-(\kappa_R + i\kappa_I)\Delta\theta}/|A_{\mathrm{CW}}|]$ plus the second-order short-range nonlinear overlap $V_{\mathrm{rep}} e^{-2\kappa_R \Delta\theta}$, which is Eq. (6). Differentiating Eq. (6) with respect to $\Delta\theta$ gives the equilibrium force condition

$$\frac{\partial U_{\mathrm{mol}}}{\partial(\Delta\theta)} = U_0 \sqrt{\kappa_R^2 + \kappa_I^2}\, e^{-\kappa_R \Delta\theta} \sin\left[\kappa_I \Delta\theta + \varphi_0 - \arctan\left(\frac{\kappa_R}{\kappa_I}\right)\right] - 2\kappa_R V_{\mathrm{rep}} e^{-2\kappa_R \Delta\theta} = 0. \tag{15}$$

In the weak-core limit $V_{\mathrm{rep}} \to 0$, the zeros of Eq. (15) with positive second derivative $\partial^2 U_{\mathrm{mol}}/\partial(\Delta\theta)^2 > 0$ occur precisely when the sine argument equals $(2n-1)\pi$ for $n = 1, 2, 3, \ldots$, yielding Eq. (7). Finally, since the repulsive core term $-2\kappa_R V_{\mathrm{rep}} e^{-2\kappa_R \Delta\theta} \le 0$ for all $V_{\mathrm{rep}} \ge 0$, it shifts every root of Eq. (15) monotonically toward larger $\Delta\theta$, so the $n = 1$ equilibrium spacing satisfies $\Delta\theta_1^*(V_{\mathrm{rep}}) \ge \Delta\theta_1^*(0)$, which upon multiplying by $\tau_{\mathrm{rt}}/(2\pi)$ proves Corollary 1. □

B Proof of Theorem 2 (Flaring MOPA and Dispersion-Tapered Compression)

Proof of Theorem 2. In the flaring QCL MOPA, the transverse mode area scales as $A_{\mathrm{eff}}(z) = A_{\mathrm{eff}}(0) e^{\alpha_{\mathrm{tap}} z}$, so the local saturation fluence is $E_{\mathrm{sat}}(z) = A_{\mathrm{eff}}(z)\hbar\omega_0/(\Gamma q^2 z_{32}^2/\varepsilon_0) = E_{\mathrm{sat}}(0) e^{\alpha_{\mathrm{tap}} z}$. The Frantz–Nodvik energy transport equation for the pulse energy $E(z) = \int |U(z,t)|^2 \, \mathrm{d}t$ along the flaring amplifier is

$$\frac{\mathrm{d}E(z)}{\mathrm{d}z} = g_{\mathrm{MOPA}} E_{\mathrm{sat}}(z) \left[1 - \exp\left(-\frac{E(z)}{E_{\mathrm{sat}}(z)}\right)\right] - \alpha_{\mathrm{MOPA}} E(z). \tag{16}$$

Define the normalized fluence variable $\Phi(z) \equiv E(z)/E_{\mathrm{sat}}(z) = E(z) e^{-\alpha_{\mathrm{tap}} z}/E_{\mathrm{sat}}(0)$. Differentiating $\Phi(z)$ with respect to z and substituting Eq. (16) yields the autonomous separable differential equation

$$\frac{\mathrm{d}\Phi}{\mathrm{d}z} = g_{\mathrm{MOPA}}\left(1 - e^{-\Phi}\right) - \left(\alpha_{\mathrm{MOPA}} + \alpha_{\mathrm{tap}}\right)\Phi \approx \left(g_{\mathrm{net}} - \alpha_{\mathrm{tap}}\right)\left(e^{\Phi} - 1\right) e^{-\Phi}. \tag{17}$$

Multiplying both sides by $e^{\Phi}/(e^{\Phi} - 1)$ and integrating from $z = 0$ ($\Phi(0) = E_{\mathrm{in}}/E_{\mathrm{sat}}(0)$) to $z = L_{\mathrm{MOPA}}$ gives $\ln(e^{\Phi(L_{\mathrm{MOPA}})} - 1) - \ln(e^{\Phi(0)} - 1) = (g_{\mathrm{net}} - \alpha_{\mathrm{tap}}) L_{\mathrm{MOPA}}$, which solves immediately for $E_{\mathrm{MOPA}}(L_{\mathrm{MOPA}}) = E_{\mathrm{sat}}(0) e^{\alpha_{\mathrm{tap}} L_{\mathrm{MOPA}}} \Phi(L_{\mathrm{MOPA}})$, proving Eq. (10).

For Part (b), in the passive $\mathrm{In}_{0.53}\mathrm{Ga}_{0.47}\mathrm{As}$ waveguide (9) with $\beta_3 = 0$, the pulse energy evolves as $E_{\mathrm{comp}}(z) = E_{\mathrm{MOPA}}(L_{\mathrm{MOPA}}) e^{-\alpha_{\mathrm{comp}} z}$. Introducing the rescaled dispersion coordinate $\zeta(z) = \int_0^z |\beta_2(z')| \, \mathrm{d}z'$ and field transformation $U(z,t) = \sqrt{|\beta_2(z)|/\gamma_{\mathrm{NL}}}\, V(\zeta, t)$ transforms Eq. (9) into the exact integrable NLSE whenever σ_{tap} matches the adiabatic invariant $E_{\mathrm{comp}}(z) T(z)/|\beta_2(z)| = 2/\gamma_{\mathrm{NL}}$. Solving this invariant for the FWHM duration $\tau_{\mathrm{ad}}(z) = 1.763\, T(z)$ yields the compression factor $\mathscr{F}_{\mathrm{comp}}(z) = \tau_0/\tau_{\mathrm{ad}}(z)$ in Eq. (11). Finally, retaining the third-order dispersion perturbation $\frac{1}{6}\beta_3 \partial_t^3 U$ in the collective pulse-width momentum integral $\frac{\mathrm{d}^2}{\mathrm{d}z^2} \int t^2 |U|^2 \, \mathrm{d}t$ adds a positive repulsive dispersive potential $\propto \beta_3^2/T^4$ that halts compression when the third-order dispersion length $L_{D3} = T_{\mathrm{min}}^3/|\beta_3|$ equals twice the nonlinear length $L_{\mathrm{NL}} = (\gamma_{\mathrm{NL}} P_{\mathrm{peak}})^{-1}$, giving $T_{\mathrm{min}} = (2|\beta_3|/[\gamma_{\mathrm{NL}} P_{\mathrm{peak}}])^{1/3}$ and proving Eq. (12). □

C Jacobi Variational Stability of Soliton Molecules

Perturbing the n-th equilibrium molecule spacing $\Delta\theta_n^*$ by $\delta\theta(\tau) = \Delta\theta(\tau) - \Delta\theta_n^*$ and relative phase $\delta\psi(\tau) = \psi_2(\tau) - \psi_1(\tau)$ in the coupled two-soliton collective coordinate equations yields the 2×2 Jacobian matrix

$$\mathscr{J}_n = -\begin{pmatrix} \partial^2 U_{\mathrm{mol}}/\partial(\Delta\theta)^2|_{\Delta\theta_n^*} & \kappa_I U_0 e^{-\kappa_R \Delta\theta_n^*}\sin(\varphi_0) \\ -\kappa_I U_0 e^{-\kappa_R\Delta\theta_n^*}\sin(\varphi_0) & \kappa_R U_0(\kappa_R^2+\kappa_I^2)^{1/2} e^{-\kappa_R\Delta\theta_n^*} \end{pmatrix}. \tag{18}$$

At the odd stationary branches $\Delta\theta_n^*$ ($n = 1,2,3,\dots$) of Eq. (7), $\partial^2 U_{\mathrm{mol}}/\partial(\Delta\theta)^2|_{\Delta\theta_n^*} = U_0(\kappa_R^2+\kappa_I^2)e^{-\kappa_R\Delta\theta_n^*} > 0$, so $\mathrm{Tr}(\mathscr{J}_n) < 0$ and $\det(\mathscr{J}_n) > 0$, guaranteeing Hurwitz exponential stability of the bound soliton molecules against both spacing and relative-phase perturbations.

Data and Code Availability

All simulation scripts, standalone TikZ figure generators, and JSON verification datasets needed to reproduce every calculation and figure in this paper are provided in `/papers/paper_3_topological_soliton_molecules/`:

- `generate_paper3_data_and_figures.py`: Self-contained Python simulation and vector PDF figure generator implementing Theorem 1, the 2,000-cavity verification of Corollary 1, Theorem 2 MOPA + compressor evolution, and standalone TikZ PDF compilation.
- `reproduce_paper3.ts`: TypeScript verification module reproducing the phase-kink bifurcation window, quantized molecule spacings, and MOPA compression laws.
- `results/`: JSON datasets (`theorem1_kink_and_molecules.json`, `corollary1_2000_molecules.json`, `theorem2_mopa_compression.json`, and `spatiotemporal_soliton_check.json`).
- `figures/`: Standalone vector PDF figures (`fig1_chip_and_argand_kink.pdf` through `fig5_compressed_pulse_and_molecule.pdf`) and their LaTeX/TikZ source files.

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